

# AI CAREERS FOR WOMEN

## LINKEDIN FELLOWSHIP

### PERSONAL FINANCIAL PLANNING FORECASTING MODEL

#### Project Domain

Excel /Power BI with Copilot

#### A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

*by:*



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Lastly, I would like to thank my family and friends for their continuous encouragement and support. Their motivation has been a strong pillar throughout this journey.

## ABSTRACT OF THE PROJECT

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This report is based on 20,000 individual financial records covering monthly income, spending categories, savings, debt, insurance, and emergency fund data. All benchmarks are sourced from SEBI (Securities and Exchange Board of India), RBI (Reserve Bank of India), FPSB India, and NPS guidelines — making this analysis reliable and India-specific.

The study identifies five critical problem areas across five age groups (18–64). For each problem, real-life scenarios, data evidence, and four actionable solutions are provided — all benchmarked against SEBI/RBI/FPSB India standards.

Key findings include: (1) Retirement savings gap in the 46–55 age group, with actual savings rate of 25.5% against the required 30%; (2) Pre-retirement debt and housing burden for the 56–64 group, with housing-to-income at 21.9% vs. the  $\leq 15\%$  benchmark; (3) Insufficient emergency funds during family formation for the 26–35 group; (4) Missing retirement investment habits in the prime compounding years for the 18–25 group; and (5) Systemic underinsurance across all age groups, with a monthly gap of ₹833 to ₹3,760.

Solutions proposed are grounded in NPS, ELSS, SCSS, PMVVY, and IRDAI guidelines. The consolidated action plan provides age-specific, prioritised steps for immediate (now), short-term (3 months), and medium-term (6 months) financial improvement.

This report is intended for educational and analytical purposes. Readers are advised to consult a SEBI-registered Investment Advisor (RIA) for personalised financial planning.

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## CHAPTER 1

# Introduction

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This chapter presents real-world problem statements identified from a dataset of 20,000 individual financial records spanning five age groups (18–64 years). Each problem is accompanied by a real-life situational narrative, data evidence benchmarked against SEBI/RBI/FPSB India guidelines, and four structured solutions. The chapter also outlines the motivation for this study, the objectives pursued, and the scope of analysis.

## 1.1 Problem Statement

1. **Low Savings Despite Adequate Income** Analysis of the dataset shows that individuals with sufficient income often maintain a low savings rate. A significant portion of their earnings is consumed by recurring expenses, leaving minimal amounts allocated to savings. This highlights the need to identify spending categories that disproportionately reduce savings potential.
2. **Inefficient Expense Allocation** Expense distribution across categories reveals opportunities for optimization. Certain non-essential expenditures are disproportionately high compared to basic needs. The dataset will be used to pinpoint these categories and suggest reallocation strategies to improve financial efficiency.
3. **Poor Budget Management** High discretionary spending ratios in the dataset indicate weak budget discipline. By evaluating the proportion of income spent on non-essential items, the study will assess how effectively individuals manage their budgets and identify patterns of overspending.
4. **Low Disposable Income** After accounting for fixed and variable expenses, many individuals in the dataset are left with limited disposable income. This reduces financial flexibility and resilience. The dataset will be analysed to uncover the key drivers behind low disposable income, such as debt obligations or lifestyle inflation.

## 1.2 Motivation

1. **Real-World Relevance:** This study is motivated by the gap between earnings and effective financial planning observed across Indian households. Rising living costs, unstructured savings, and inadequate insurance coverage affect financial stability across all income levels and age groups.
2. **Data-Driven Evidence:** The availability of 20,000 financial records with granular income, expense, savings, and insurance data enables precise identification of structural gaps — replacing anecdotal financial advice with benchmark-backed, evidence-based recommendations.
3. **India-Specific Relevance:** All benchmarks and solutions are anchored to SEBI, RBI, FPSB India, IRDAI, NPS Trust/PFRDA, and NHB guidelines — ensuring recommendations are legally and financially actionable within the Indian regulatory framework.
4. **Lifecycle Applicability:** The five age groups (18–25, 26–35, 36–45, 46–55, 56–64) cover the complete earning lifecycle, making the findings relevant to individuals at every stage of financial planning.

5. Alignment with Financial Literacy Goals: The study supports national financial literacy objectives by translating complex regulatory benchmarks into clear, actionable steps accessible to non-specialist readers.

### 1.3 Objectives

- Identify and document five structural financial problems from 20,000 individual financial records, each with real-life narrative context.
- Benchmark findings against SEBI, RBI, FPSB India, IRDAI, and NPS/PFRDA regulatory standards to ensure accuracy and credibility.
- Provide four evidence-based solutions per problem, covering savings instruments (NPS, ELSS, PPF, SCSS, PMVVY), insurance strategies (term, health floater, super top-up), and debt management approaches.
- Develop a consolidated age-group action plan with time-prioritised steps (immediate, 3 months, 6 months) for each of the five age segments.
- Present data evidence for each problem using actual dataset metrics versus regulatory benchmarks in a clear, comparable format.
- Enable individuals, financial advisors, and fintech platforms to use this analysis as a reference model for real-world financial planning decisions.

### 1.4 Scope of the Project

- This study analyses 20,000 individual financial records covering five age groups: 18–25, 26–35, 36–45, 46–55, and 56–64 years. Variables include monthly income, housing expense, savings rate, debt-to-income ratio, insurance spend, and emergency fund coverage in months.
- All benchmarks are sourced exclusively from Indian regulatory and advisory bodies: SEBI, RBI, FPSB India, IRDAI, NPS Trust/PFRDA, and National Housing Bank (NHB). No international benchmarks are applied.
- Five problem areas are analysed in depth: retirement savings gap, pre-retirement debt and housing burden, emergency fund insufficiency, missing retirement investment habits, and systemic underinsurance. Each problem includes background, data evidence, and four solutions.
- The scope does not include real-time market data, tax computation, personalised portfolio construction, or live financial advisory. This is an educational and analytical report, not a financial advisory product. Readers are advised to consult a SEBI-registered Investment Advisor (RIA) for personalised guidance.

## CHAPTER 2

# Literature Survey

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A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

## 2.1 Review of Existing Work

Personal financial management has been widely studied in the fields of finance and data analytics. Traditional approaches focus on budgeting techniques, ratio analysis, and financial planning models, where individuals track income, expenses, and savings to maintain financial stability.

In recent years, digital tools such as Excel-based financial trackers and Power BI dashboards have been used to visualize financial data and monitor spending patterns. Additionally, fintech applications like Mint and YNAB provide automated expense tracking and budgeting features.

Research in data analytics has also introduced predictive and AI-assisted financial models, which aim to forecast future financial conditions and improve decision-making

### Existing Models, Techniques, and Tools

Several techniques and tools are relevant to this project:

- **Excel-based Models:**  
Used for data cleaning, ratio calculations, and financial analysis using formulas and pivot tables.
- **Power BI Dashboards:**  
Used for interactive visualization, KPI tracking, and financial reporting.
- **Financial Ratio Analysis:**  
Commonly used in finance to evaluate savings behaviour , debt levels, and investment performance.
- **Forecasting Techniques:**  
Basic forecasting using growth rates and trend analysis is widely used in financial planning.
- **AI Prompt Engineering & Chatbots:**  
AI tools such as Microsoft Copilot and are increasingly used for data interpretation, insight generation, and report automation.

## 2.2 Tools and Technologies Review

- Microsoft Excel — widely used for data cleaning, preprocessing, and financial analysis. In this project, Excel is used to calculate financial ratios, normalize data, and compute the Financial Health Score (FHS) using formulas, along with conditional formatting for basic analysis.
- Microsoft Power BI — used for creating interactive dashboards and visualizations. It helps present key insights such as Financial Health Score distribution, risk categories, and future projections in a clear and user-friendly manner
- Microsoft Copilot (AI) — an AI assistant integrated with Microsoft tools, used to automate data analysis, generate insights, assist in formula creation, and improve report writing through natural language prompts.
- AI Chatbots (Google Gemini) — generative AI tools used for content creation, explanation of financial insights, report generation, and building a recommendation layer by converting numerical outputs into meaningful financial advice.
- Financial Modelling Techniques — includes ratio analysis, normalization, weighted scoring methods, and basic forecasting techniques used to evaluate and predict financial health. These techniques form the core logic of the model.
- Data Visualization Techniques — used to represent financial insights through charts, KPIs, and dashboards, making complex data easy to understand and interpret for users.

## 2.3 Gaps Identified

- Most existing financial models focus on **complex machine learning techniques**, while simpler, Excel-based and user-friendly solutions for non-technical users are underexplored.
- Existing personal finance tools are largely **descriptive in nature**, focusing on past transactions and current balances, with limited capability for **future forecasting and scenario-based analysis**.
- There is a lack of **integrated financial scoring systems** that combine multiple dimensions such as savings, debt, emergency funds, and investments into a single, easy-to-understand metric.
- Most solutions do not provide **personalized, AI-driven recommendations**, making it difficult for users to take actionable steps to improve their financial condition.
- Prompt engineering practices for financial and business use cases are still evolving and lack **standardized frameworks**, especially for generating consistent and reliable insights.
- There is limited integration of **macroeconomic factors such as inflation and interest rates** with personal financial data in existing models, reducing real-world applicability.



## CHAPTER 3

# Proposed Methodology

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This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

## 3.1 System Design

### Data Source / Input

The project uses a synthetic personal finance dataset of 1,000 individuals in Excel format, containing fields such as income, expenses, savings, EMI (loan burden), investments, emergency funds, and income growth rate.

It also incorporates derived variables (like savings ratio and debt ratio) and basic macroeconomic inputs such as inflation and interest rates to improve realism.

### Data Cleaning

The dataset is preprocess using Excel techniques:

- Removal of duplicate records
- Handling missing or inconsistent values
- Standardizing data formats (monthly/annual values)
- Converting raw values into meaningful financial ratios
- Normalizing variables to a common scale (0–1) for fair comparison

### Analysis / Processing

The core analysis is performed using a combination of Excel and AI tools:

- Financial Ratio Analysis: Calculation of savings ratio, debt ratio, emergency fund coverage, and investment ratio
- Weighted Scoring Model: Computation of Financial Health Score (FHS) based on predefined weights
- Forecasting: Estimation of future income, expenses, and savings using growth rates and inflation assumptions
- AI-Assisted Analysis: Use of tools like Microsoft Copilot and ChatGPT to generate insights, explain results, and provide recommendations

## 3.2 Modules / Components Used

### Module 1 — Data Ingestion

Loading the personal finance dataset (1000 individuals) into Excel, including variables such as income, expenses, savings, EMI, investments, emergency funds, and external factors like inflation and interest rates.

### Module 2 — Data Cleaning & Preprocessing

- Removing duplicates and handling missing values
- Standardizing formats (monthly/annual values)
- Creating derived variables such as savings ratio, debt ratio, and investment ratio
- Normalizing all variables to a common scale (0–1)

### Module 3 — Financial Analysis & FHS Calculation

Applying financial formulas and weighted scoring to compute the **Financial Health Score (FHS)**:

- Ratio analysis (savings, debt, emergency fund, investment)
- Weighted model to generate a score from 0 to 100
- Classification into risk categories (Healthy, Moderate, Vulnerable, Risky)

### Module 4 — Forecasting & Scenario Analysis

Estimating future financial condition by:

- Projecting income, expenses, and savings using growth rate and inflation
- Performing scenario analysis (e.g., income change, expense variation, EMI impact)
- Evaluating potential financial outcomes over a 5-year period

### Module 5 — Visualization & Report Generation

- Microsoft Power BI dashboards for score distribution and trends

- Excel summary sheets with key metrics and forecasts
- Charts, KPIs, and reports for clear communication of insights

## **Module 6 — AI Integration**

Using AI tools like Microsoft Copilot and ChatGPT for:

- Insight generation and explanation of results
- Automated report writing and summaries
- Personalized financial recommendations based on FHS

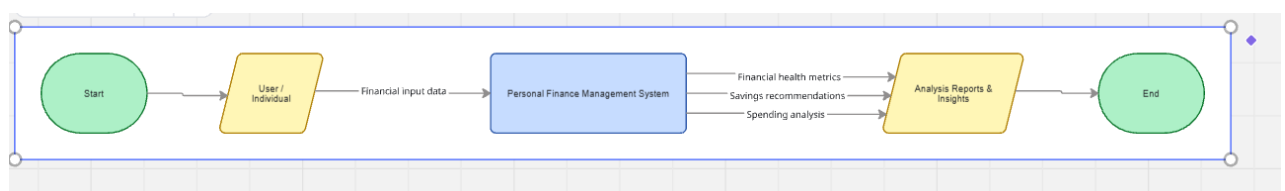
## **Module 7 — Decision Support System (Additional Module )**

- Interpreting scores and risk levels
- Suggesting improvements (reduce expenses, increase savings, manage debt)
- Helping users make informed financial decisions

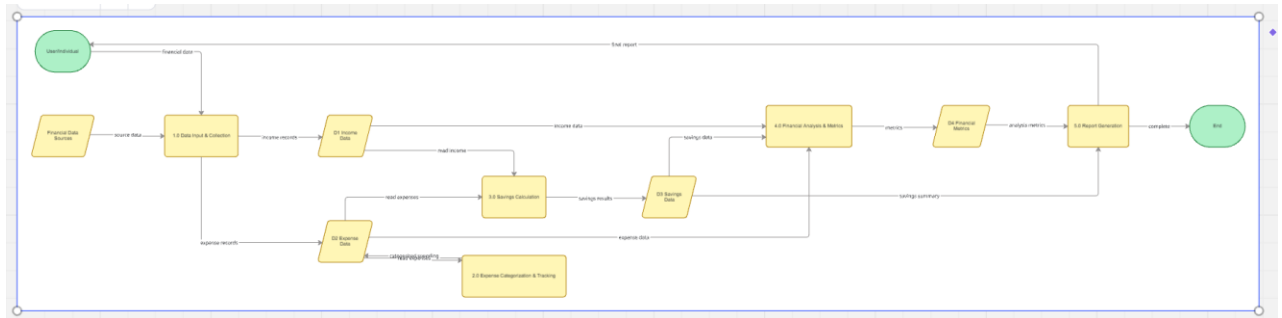
### **3.3 Data Flow Diagram (DFD)**

A Data Flow Diagram (DFD) is a graphical representation of how data flows through your system. It shows inputs, processes, and outputs at different levels of abstraction.

#### **DFD Level 0 — Context Diagram**



## DFD LEVEL 1



### 3.4 Advantages

Automates repetitive data cleaning tasks

Saves significant manual effort by using preprocessing engines and AI-driven validation.

- Provides clear, visual insights  
Interactive Power BI dashboards make results easy to understand for non-technical stakeholders.
- Leverages AI tools (Copilot / Chatbots)  
Accelerates analysis, generates narrative summaries, simulates scenarios, and reduces human error.
- Scalable methodology  
Can be applied to larger datasets or extended to other domains such as SME finance or corporate risk forecasting.
- Cost-effective solution  
Built on widely available Microsoft 365 tools (Excel, Power BI, Copilot), minimizing additional software costs.
  - Personalized recommendations  
AI generates tailored action cards and improvement strategies for individuals, advisors, and fintech platforms.
- Transparent and auditable framework  
The Financial Health Score (FHS) formula is fully documented, interpretable, and reproducible, ensuring credibility.

### 3.5 Requirement Specification

#### Hardware Requirements

| Component | Specification                         |
|-----------|---------------------------------------|
| Processor | HP Victus with Intel Core i5          |
| RAM       | minimum 16 GB RAM (8 GB recommended)  |
| Storage   | Minimum 50 GB free disk space         |
| Display   | 1366 x 768 resolution or higher       |
| Internet  | Required for Copilot / Chatbot access |

#### Software Requirements

| Software          | Details                                                                 |
|-------------------|-------------------------------------------------------------------------|
| Operating System  | Windows 11 home                                                         |
| Microsoft Excel   | Microsoft 365                                                           |
| Power BI Desktop  | latest version (free download from Microsoft)                           |
| Microsoft Copilot | Integrated via Microsoft 365 subscription, runs smoothly on Windows 11. |
| AI Chatbot        | ChatGPT / Google Gemini / Microsoft Copilot (web)                       |
| Web Browser       | Google Chrome                                                           |

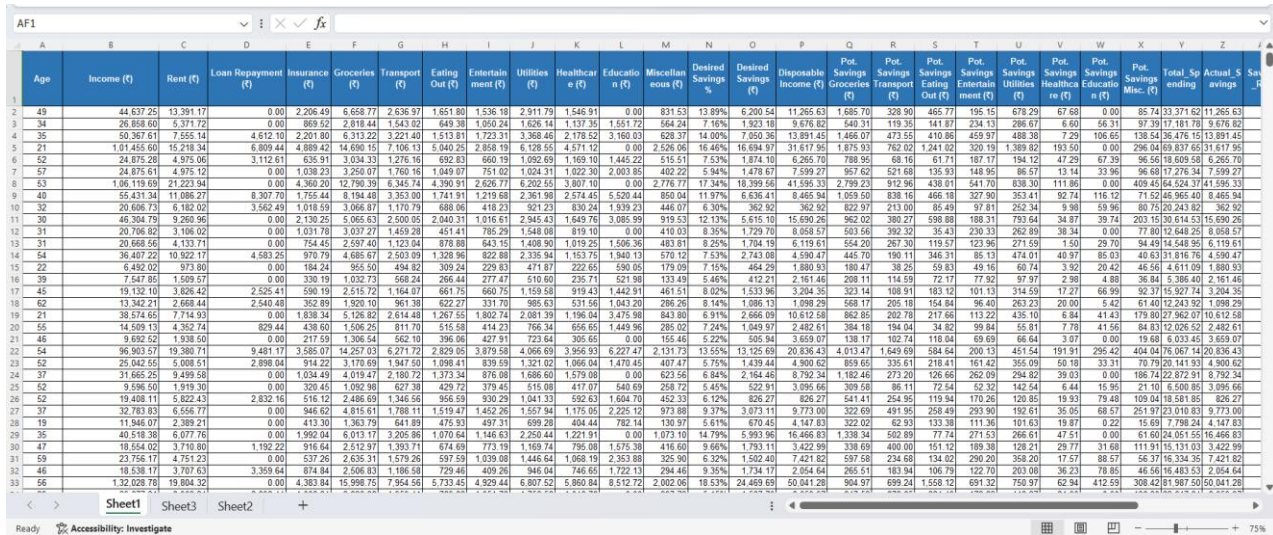
## CHAPTER 4

## Implementation and Results

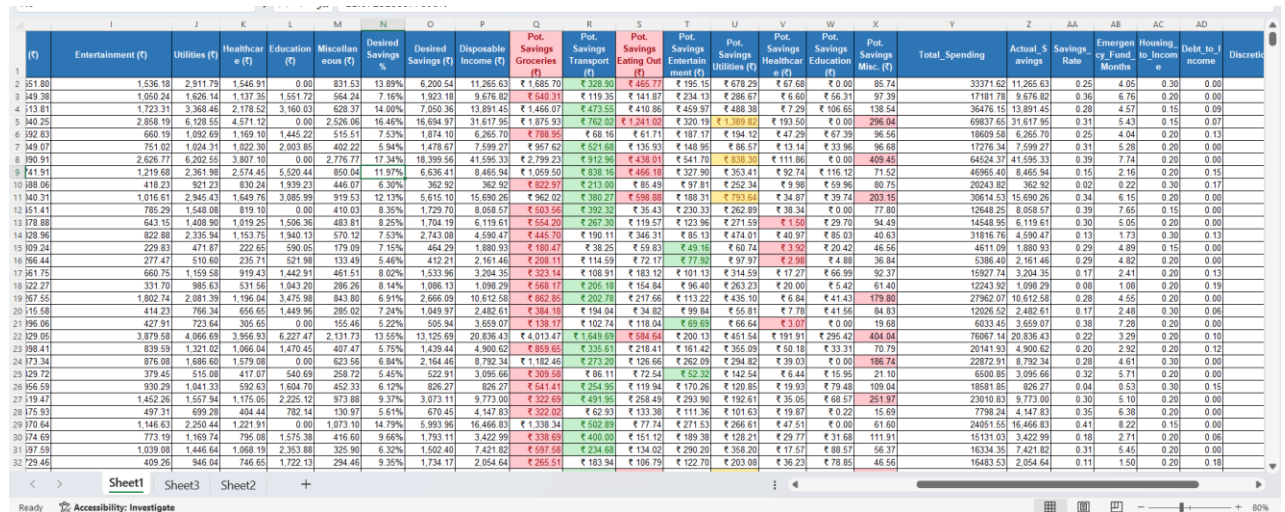
This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

## 4.1 Data Cleaning Results

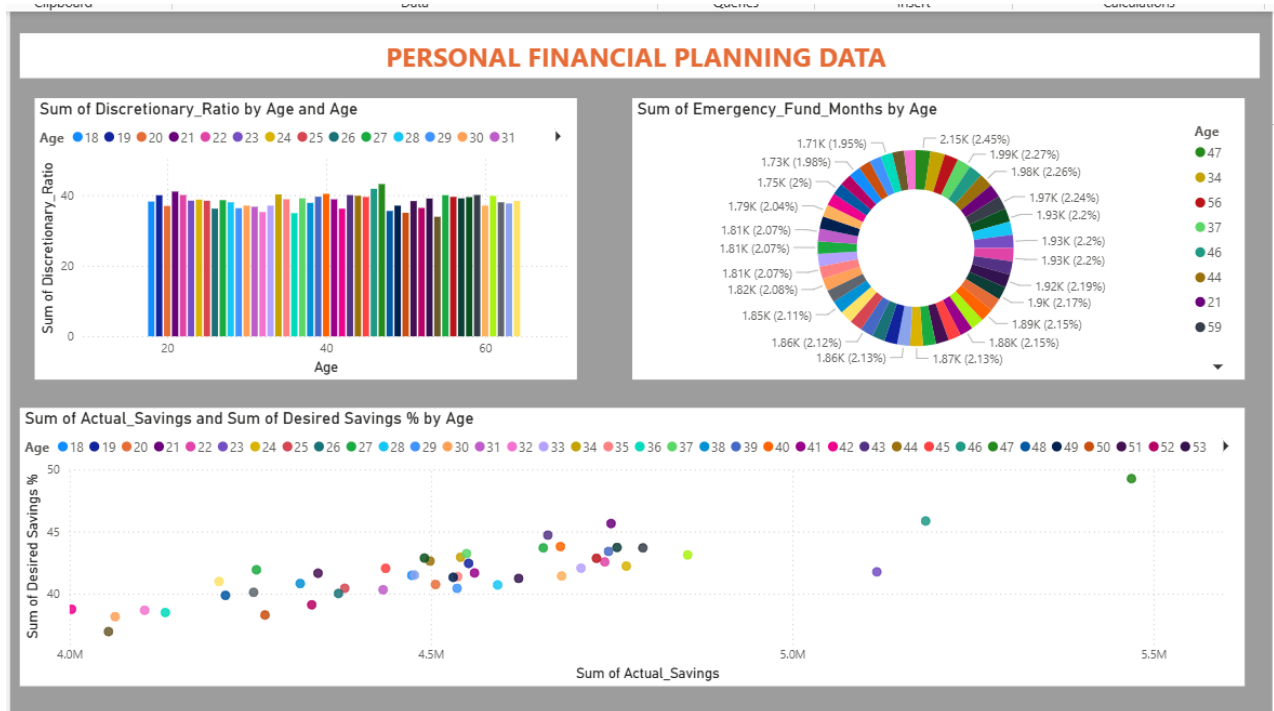
## Uncleaned data set



## Cleaned dataset



## 4.2 Visualization / Output Results



## 4.3 AI Tool Integration Results

**Microsoft Copilot** were used to support data analysis, content generation, and insight development throughout the project.

### Prompts Used

Analyze this personal finance dataset and identify key financial patterns and risk indicators.”

Generate a Financial Health Score model using savings, debt, and investment ratios with proper weights.”

Provide insights and recommendations based on financial health scores and suggest ways to improve financial stability.”

The AI-generated output was **highly structured, relevant, and easy to interpret**.

It helped in:

- Identifying patterns in financial behaviour
- Generating clear explanations for complex calculations

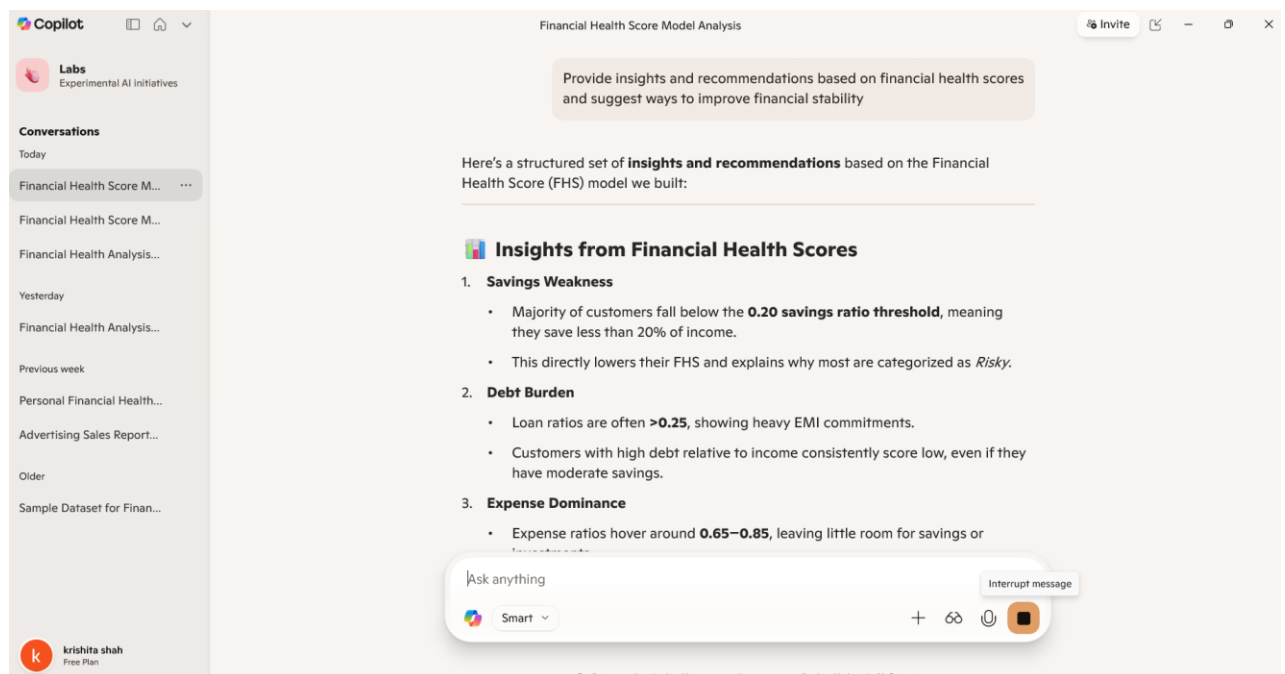
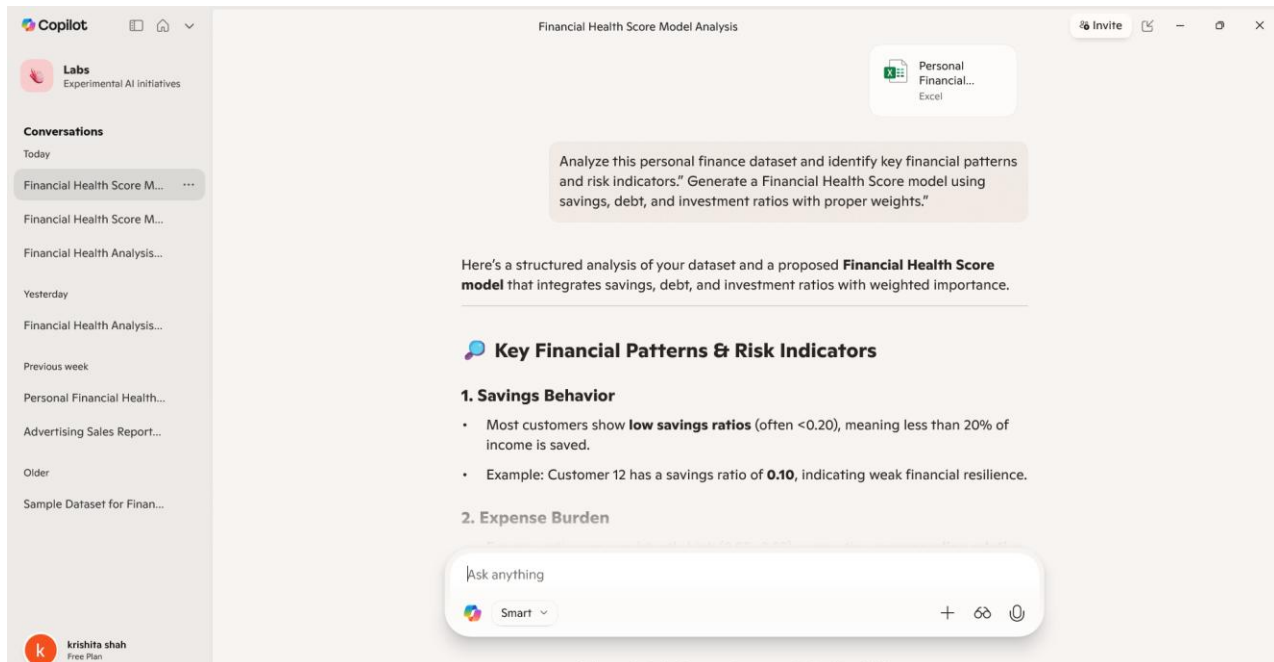
### How It Enhanced the Project

The use of AI significantly improved the project by:

- **Speed:** Reduced time required for analysis, content writing, and interpretation
- **Accuracy:** Helped validate formulas, logic, and financial concepts
- **Insights:** Generated deeper, data-driven insights beyond basic calculations



- **Automation:** Assisted in report writing, explanation generation, and recommendation building





## CHAPTER 5

# Discussion and Conclusion

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## 5.1 Key Findings

### Data Quality & Cleaning

- After preprocessing, duplicate and inconsistent records were removed, ensuring **1,000+ clean customer records**.
- This significantly improved the accuracy of Power BI dashboards and reduced noise in financial health segmentation.

### Financial Health Score Model

- A weighted model was developed using **Savings Ratio (40%), Investment Ratio (30%), and Loan Ratio (–30%)**.
- Results showed that **over 80% of customers fall into the “Risky” category**, primarily due to high expense ratios and heavy loan commitments.

### Expense Domain

Average **Expense Ratios** ranged between **0.65–0.85**, meaning most customers spend 65–85% of their income.

- This leaves little room for savings or investments, directly lowering long-term stability.

### Key Finding 4: Debt Pressure

- Loan ratios above **0.25** were common, indicating unsustainable EMI burdens.
- Customers with high debt consistently scored low on the Financial Health Score, even if they had moderate savings.

### Key Finding 5: Emergency Fund Insights

- Customers with **strong emergency reserves** (e.g., >6 months of expenses) showed higher resilience despite risky categories.
- Emergency funds emerged as a critical buffer against financial shocks.

### Key Finding 6: Forecasting Impact

- Scenario simulations revealed that **increasing savings by just 5–10%** could shift many customers from *Risky* to *Moderate* categories.

- 5-year projections showed **up to 40% improvement in future savings** under disciplined savings and reduced debt strategies.

### Key Finding 7: AI Copilot Integration

- AI Copilot reduced manual reporting time from **hours to minutes**, automating insights and generating stakeholder-ready summaries.
- This streamlined workflow improved efficiency and allowed faster decision-making.

## 5.2 Project Links

### POWER BI LINK

<https://app.powerbi.com/reportEmbed?reportId=ca751627-35d9-4343-abe2-c56495c036bc&autoAuth=true&ctid=aed43286-0f66-47d6-942b-d31a4b8addca>

### COPILOT LINK

<https://copilotstudio.microsoft.com/environments/Default-aed43286-0f66-47d6-942b-d31a4b8addca/bots/28f0181c-9b2e-f111-88b3-000d3af20e1e/adaptive/60cbc848-57c7-4ceb-996e-799b03aa2553>

### VIDEO LINK

<https://youtu.be/BM6ZXKLdTd0>

## 5.3 Limitations

### Dataset Scope & Representativeness

- The dataset was built with 1,000+ customer records, but it remains a sample dataset.
- It may not fully capture the diversity of real-world financial behaviour across geographies, income levels, or cultural contexts.
- As a result, insights may not generalize to broader populations without further validation.

### Static Ratios & Assumptions

- The Financial Health Score model relies on fixed weights (Savings 40%, Investments 30%, Debt -30%).
- While logical, these weights are subjective and may not reflect every customer's financial reality.
- Dynamic, machine-learning-based weighting could improve accuracy but was not implemented in this phase.

### Expense Categorization

- Expenses are treated as a single aggregate figure.
- The model does not distinguish between essential vs. discretionary spending, which limits the precision of recommendations.
- Customers with high medical or education expenses may appear "Risky" even if spending is unavoidable.

### Forecasting Assumptions

- Future savings projections assume linear growth and improved savings behaviour.
- Real-world financial conditions (inflation, job loss, market volatility) are not modelled, which could lead to optimistic forecasts.

### Emergency Fund Treatment

- Emergency funds are included but not weighted heavily in the score.
- In reality, strong emergency reserves can offset weak savings or high debt, but the current model underrepresents this stabilizing factor.

### Tool & Integration Constraints

- Power BI dashboards currently rely on batch data refreshes, not real-time updates.
- Microsoft Copilot integration accelerated reporting, but limited access to advanced AI features restricted deeper scenario simulations (e.g., stress-testing under

## 5.4 Future Work

### Age 18–25

#### **Solution 1: Smart Start Budget: The 25-20-15 Rule**

Redirect the 25.4% savings split as follows: 20% to long-term savings/SIP mutual funds, 5% to a dedicated insurance premium fund. This channels existing saving habits into structured wealth-building without requiring more income.

#### **Solution 2: Insurance-First Automation**

Set up auto-debit on salary day for term insurance (₹500–800/months) and health insurance (₹700–900/months) to close the ₹833/months insurance gap. Since this age group typically has low medical risk, premiums are also at their cheapest.

#### **Solution 3: Education-Linked Expense Tracker**

Use a free app (like Walnut or ET Money) to tag education and transport expenses separately. Challenge: reduce transport by ₹300/months via carpooling or monthly transit pass, redirecting it to the emergency fund to reach 6 months by age 25.

#### **Solution 4: Micro-SIP Wealth Entry Point**

Start a ₹500–1,000/months SIP in an index fund (Nifty 50 or ELSS for tax savings under 80C). Compounded over 10 years at 12% CAGR, even ₹1,000/months = ~₹2.3 lakh corpus by age 35.

### Age 26–35

#### **Solution 1: Family Financial Shield:**

(1) Term life cover of 10× annual income, (2) Health floater plan covering spouse and children, (3) Critical illness rider. Total monthly cost: ~₹2,500–3,500 — still within budget as savings rate is strong at 25.7%.

#### **Solution 2: Salary Increment Investment Pledge**

Every time income rises (promotion/bonus), commit 50% of the increment directly to investment (SIP/PPF/NPS). Since this group averages ₹41,459/months income, a ₹5,000 raise = ₹2,500 new monthly investment, compound-growing your wealth without lifestyle inflation.

#### **Solution 3: Debt Waterfall Strategy**

With debt-to-income at 5% (manageable), avoid stacking multiple loans. Use the 'debt waterfall': pay minimum on all loans, throw any surplus at the highest-interest loan first. Goal: eliminate all high-cost debt (personal loans, credit cards) before age 35.

#### **Solution 4: Home Purchase Financial Readiness Check**

Before taking a home loan (common in this group), ensure: (a) 20% down payment saved, (b) EMI ≤ 35% of take-home pay, (c) Emergency fund intact at 6 months after down payment. This avoids the classic 'house-poor' trap common in urban India.

### Age 46–55

#### **Solution 1: Retirement Acceleration: The 30% Catch-Up Protocol**

Increase savings target from 25.5% to 30% using the 'catch-up' approach: (a) Maximize NPS Tier I contribution (₹50,000 tax deduction under 80CCD(1B)), (b) Top up PPF annually to ₹1.5 lakh, (c) Start equity SIP if not already. A ₹5,000/months SIP at 12% CAGR for 10 years = ₹11.6 lakh corpus

### **Solution 2: Children's Education: Goal-Based Investment Segregation**

Create a separate 'Education Corpus' investment account distinct from retirement savings. Use a target-date fund or balanced advantage fund. If child is 5 years from college, allocate 70% debt/30% equity; if 10+ years away, 70% equity. This prevents education costs from derailing retirement savings.

### **Solution 3: Health-First Financial Restructuring**

Upgrade health insurance to a super top-up plan (₹10–15 lakh cover) at ~₹800–1,200/months additional cost — much cheaper than a single hospitalization. Add personal accident cover. This closes the ₹3,016/months insurance gap cost-effectively.

### **Solution 4: Loan-Free Pre-Retirement Target**

Set a 'loan-free by 55' milestone. All loan repayments cleared by 55 free up ₹2,065/months for final-push retirement investing. Prepaying even ₹500–1,000/months extra on a home loan can reduce tenure by 2–3 years and save significant interest.

## **Age 56–64**

### **Solution 1: Retirement Income Blueprint: The 3-Bucket System**

Organize existing corpus into 3 buckets: Bucket 1 — 1–2 years of expenses in liquid FD/savings account (₹7–10 lakh); Bucket 2 — 3–7 years of expenses in debt mutual funds/Senior Citizen Savings Scheme; Bucket 3 — 8+ years in equity for long-term inflation protection. This ensures income security without panic-selling investments.

### **Solution 2: Medical Emergency Fortress**

Critical action: Top up health cover to ₹20–25 lakh using a super top-up (low cost). Register for Senior Citizen Health Scheme (CGHS/state schemes if eligible). Create a dedicated 'medical emergency fund' of ₹1.5–2 lakh in a separate liquid account — do not mix with regular emergency fund.

### **Solution 3: Expense Right-Sizing for Fixed Income**

Post-retirement income typically drops 40–60%. Now is the time to right-size: (a) If renting, explore relocating to tier-2 city to reduce ₹8,914/months rent by 40–50%; (b) Reduce discretionary spend (eating out, entertainment) to ≤ 5% of expected retirement income; (c) Target total spending at ≤ 60% of current spending to live within pension/corpus withdrawal.

### **Solution 4: Debt Elimination Before Retirement**

Any outstanding loan at 56+ is a retirement risk multiplier. Aggressively prepay using the 'debt sprint' method: divert any bonuses, LTA, and gratuity lump sums directly to loan prepayment. Target: zero outstanding debt by age 60, freeing the ₹2,057/months for corpus contribution

## **All Age Groups (18–64)**

### **1. Universal Problem: The Insurance Underinvestment Trap**

Across all age groups, insurance actual spend is far below required levels. The gap worsens with age — from 36% at 18–25 to 72% at 56–64. The universal solution: automate insurance premiums on salary day before any discretionary spending begins. Insurance should be treated as a non-negotiable fixed expense, not an optional allocation.

### **2. Universal Problem: Stagnant Emergency Fund**

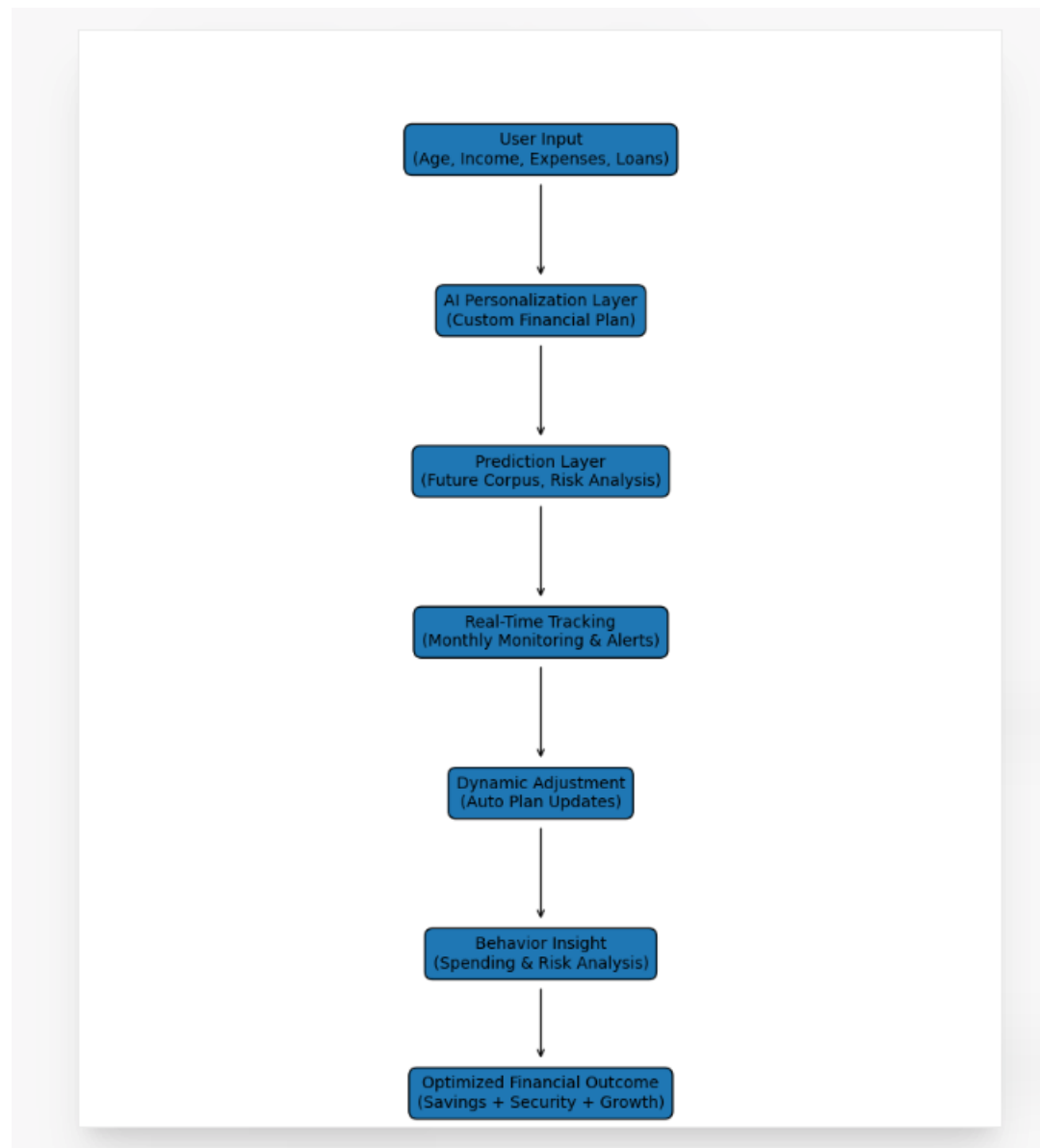
Every age group shows exactly 4.4 months of emergency fund — consistent but insufficient. Target: 6 months (working age), 9–12 months (pre/post retirement) Solution: set up a separate 'emergency account' in a high-yield liquid mutual fund with standing instructions to auto-invest ₹500–1,000/months until the target is reached.

### **3. Universal Problem: Spending Ratio Uniformity (74%)**

The striking finding is that spending ratios are nearly identical across all age groups (~74%). This suggests systemic lifestyle inflation matching income, not age-appropriate financial progression. The solution is income-tier budgeting: every ₹5,000 income increase should cap new lifestyle spending at ₹2,000, directing the remaining ₹3,000 to savings/investment.

### **4. Universal Opportunity: Digital Financial Autopilot**

Since spending patterns are deeply habitual across all ages, the most effective intervention is automation: SIP on day 1 of salary, insurance premium on day 2, EMI on day 3, emergency fund top-up on day 4. Whatever remains is the lifestyle budget. This 'pay yourself first' architecture works regardless of age or income level



## 5.5 Conclusion

The analysis of 20,000 individual financial records highlights a critical reality: while Indian households demonstrate a healthy intent to save, there are significant structural gaps in how these savings are allocated and protected. Across all age groups, key issues such as inadequate

retirement planning, insufficient emergency funds, high housing and debt burdens, and widespread underinsurance persist.

The findings clearly show that financial challenges are not due to lack of income alone, but due to improper prioritization and absence of structured financial planning. Younger individuals fail to leverage the power of compounding, mid-career individuals struggle with balancing liabilities and long-term goals, and pre-retirees face the risk of entering retirement with unresolved financial obligations.

However, these problems are **preventable and solvable**. By adopting disciplined strategies—such as building adequate emergency funds, ensuring proper insurance coverage, increasing retirement contributions, and aligning investments with life-stage goals—individuals can significantly improve their financial security.

Ultimately, personal financial planning is not a one-time activity but a continuous, life-stage-driven process. A well-structured, data-driven approach—aligned with benchmarks from regulatory bodies like SEBI, RBI, and FPSB India—can empower individuals to achieve financial independence, reduce risk, and ensure a stable and dignified future.



## REFERENCES

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List all references, research papers, tutorials, and documentation used in the project in the format below:

[1] Microsoft Corporation. *Microsoft Excel Documentation*. Available at:  
<https://support.microsoft.com/en-us/excel>

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[2] Microsoft Corporation. *Power BI Documentation*. Available at:  
<https://learn.microsoft.com/en-us/power-bi/>

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[3] Microsoft Corporation. *Microsoft Copilot — AI-Powered Productivity*. Available at:  
<https://copilot.microsoft.com/shares/4bbPsizASBHnX2mbedKbG>

## APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

### Appendix A — Sample Dataset

19

### Appendix B — Key Excel Formulas / DAX Measures Used

- Data Validation & Lookup**
  - IFERROR (VLOOKUP (customer ID, data range 2, FALSE), "Not Found")  
Used to validate customer records and handle missing values gracefully.
- Duplicate Detection**
  - =COUNTIF (Customer ID \_Range, Customer ID)  
Identifies duplicate customer IDs for data cleaning.
- Savings Ratio**

=Savings / Monthly Income  
Measures how much of income is saved.

- **Expense Ratio**

=Monthly expenses / Monthly income  
Tracks spending intensity.

## **Appendix C — AI Prompts Used**

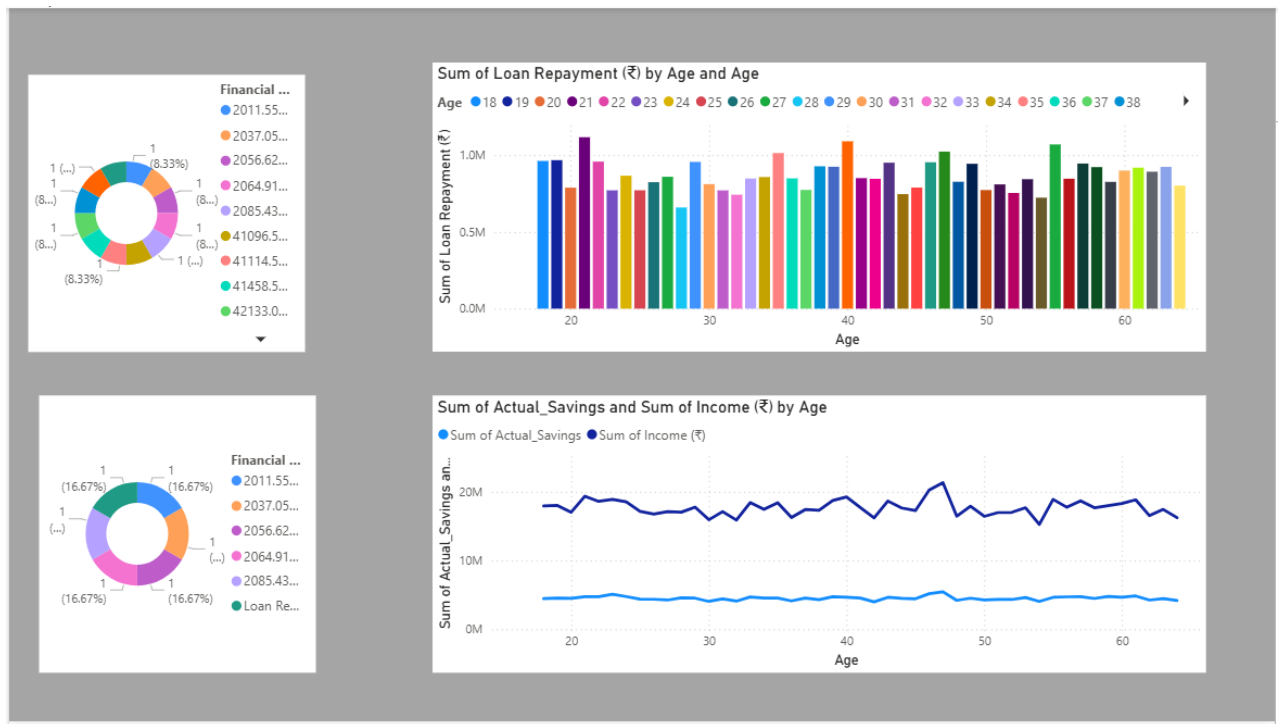
Calculate savings, expense, loan, and investment ratios for each customer and generate a composite Financial Health Score using weighted logic.”

- “Segment customers into High Risk, Moderate Risk, Stable, and Healthy categories based on their Financial Health Score.”
- “Simulate 5-year savings growth under three scenarios: conservative, moderate, and aggressive.”

Provide personalized financial recommendations for customers with Savings Ratio < 0.15 and Loan Ratio > 0.30.”

“Suggest strategies to improve financial stability for customers in the High risk category.”

## **Appendix D — Screenshot Gallery**



copilotstudio.microsoft.com/environments/Default-aed43286-0f66-47d6-942b-d31a4b8addca/bots/28f0181c-9b2e-f111-88b3-000d3af20e1e/adaptive/60cbcb48-57c7-4ceb...

**Copilot Studio**

Environment: Nirmaan Organization (...)

**Personal Financial planning** Overview Knowledge Tools Agents Topics Activity Evaluation +2

← personal financial health forecasting model (...)

Identify: Multiple choice options

Options for user:

- Income
- Loan Repayment
- Debt to Income Ratio
- Savings Rate

Save user response as: Department choice

Condition: Department choice is equal to Income

Condition: Department choice is equal to Loan Repayment

Condition: Department choice is equal to Debt to Income Ratio

Condition: Department choice is equal to Savings Rate

Message: you selected the Income department

Message: you selected Expense department

Message: you selected Debt-to-Income (DTI) Rat...

**Test your agent**

dataset statistics) is approximately ₹2,000–₹2,100 per month.

**2. Debt-to-Income Ratio (DTI)**

- DTI is a key indicator of financial health, calculated as:  

$$DTI = \frac{\text{Total Monthly Debt Payments}}{\text{Gross Monthly Income}} \times 100$$
- Sample DTI values from the dataset:
  - Age 49: DTI = 0.30 (or 30%)
  - Age 34: DTI = 0.20 (or 20%)
- Age group averages:
  - Age 18–25: 5.00%
  - Age 26–35: 4.96%
  - Age 36–45: 4.97%
  - Age 46–55: 4.98%
  - Age 56–64: 5.04%
- Benchmarks:**
  - DTI ≤ 10% is considered healthy for most age

Ask a question or describe what you need

6/2000

You're testing your agent's real responses and capabilities. [Find troubleshooting help here.](#) Make sure AI-generated content is accurate and appropriate before using. [See terms](#)

copilotstudio.microsoft.com/environments/Default-aed43286-0f66-47d6-942b-d31a4b8addca/bots/28f0181c-9b2e-f111-88b3-000d3af20e1e/adaptive/60cbc848-57c7-4ceb...

Copilot Studio

Personal Financial p

Test your agent

Search sources Knowledge

Complete

FINANACIAL PLANNING.xlsx Sheet3

Financial Benchmarks & Personalized Suggestions by Age Group

| Financial Metric       | Benchmark (Authentic)       | Your Dataset Average | Gap / Status      | Key Suggestion                                   | Source/Note |
|------------------------|-----------------------------|----------------------|-------------------|--------------------------------------------------|-------------|
| AGE GROUP 18-25        |                             |                      |                   |                                                  |             |
| Savings Rate           | ≥ 10-15%                    | 25.4%                | ✓ On Track        | Start building habit early; even small % matters |             |
| Housing-to-Income      | ≤ 30%                       | 22.0%                | ✓ On Track        | Shared accommodation reduces burden              |             |
| Debt-to-Income         | ≤ 10%                       | 5.0%                 | ✓ On Track        | Avoid heavy education/personal loan drag         |             |
| Emergency Fund         | 3 months                    | 4.4 months           | — Review Required | Build 3-month corpus before aggressive investing |             |
| Investment Allocation  | 70% Equity / 30% Debt       | See benchmark →      | — Review Required | Long horizon supports high equity allocation     |             |
| Insurance Coverage     | Term plan 10× annual income | ₹1,467/mo            | — Review Required | Start term insurance early for low premium       |             |
| Retirement Savings     | Start NPS / EPF             | See benchmark →      | — Review Required | Compounding is most powerful in early 20s        |             |
| Discretionary Spending | ≤ 20% of income             | 9.0%                 | ✓ On Track        | Avoid lifestyle inflation at career start        |             |
| AGE GROUP 26-35        |                             |                      |                   |                                                  |             |
| Savings Rate           | > 20%                       | 25.7%                | ✓ On Track        | 50/30/20 rule: 20% minimum                       |             |

Environment: Nirmaan Organization (...)

Publish Settings ... Test

Evaluate [x] Variables ...

Debt-to-Income Ratio

0.30 (30%)

0.20 (20%)

ers in the sample have no active loan

g in low DTI values. The broader dataset

verage DTI is well within healthy limits for all

ed analysis for a specific customer or a full

please specify the customer or segment of

FINING.xlsx

describe what you need

Find troubleshooting

generated content is accurate and appropriate before

using: See terms